

BHANU TEJA L

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<https://github.com/Bhanu7202> | Portfolio | Bengaluru, India

Professional Summary

Computer Science & Engineering graduate and MSc Mathematics student with practical experience in backend architecture and data processing. Proficient in Python and SQL for designing end-to-end pipelines that resolve complex data bottlenecks. Experienced in designing Big Data pipelines that handle terabyte-scale datasets. Adept at bridging the gap between analytical models and business needs by integrating REST APIs and deploying scalable, user-centric software solutions.

Work history

Data Science Intern, AI Variant

Jan 2025 – Oct 2025 | Bengaluru, India

- Engineered Python-based data pipelines using TensorFlow and PyTorch, cutting processing time by 20% and delivering accurate predictive models.
- Worked with a 5-person team to convert data insights into scalable software solutions and actionable business strategies.
- Boosted model accuracy to 85% via feature engineering; documented processes with clear, analytical rigor.

Skills

Languages & Databases — Python, SQL (MySQL), C/C++, R, HTML/CSS.

Machine Learning & Data Engineering — TensorFlow, PyTorch, Scikit-learn, Time-Series Analysis, Data Visualization, PowerBI, Tableau.

Software, Cloud & Tools — AWS, Azure, GCP, Docker, Kubernetes, CI/CD, Flask, ETL, REST APIs, Streamlit, Git/GitHub, Agile, Scrum, VS Code.

Projects

Music Recommendation Using Facial Expressions

- Designed an emotion-aware recommendation engine that maps user moods to music, improving playlist personalization.
- Built a scalable computer vision pipeline using Python, OpenCV, and CNNs (TensorFlow/ PyTorch), integrating the Spotify API to demonstrate full-stack software adaptability.
- Achieved 90% emotion classification accuracy with low-latency execution, delivering a seamless application experience that highlights technical rigor and system optimization.

Customer Churn Prediction

- Architected a telecom risk-scoring system that identifies churners and defines retention thresholds.
- Constructed robust Python pipelines to structure datasets and optimize ensemble models (LightGBM, Random Forest), utilizing A/B testing to identify the highest-performing churn predictor.
- Attained 86.5% predictive accuracy and deployed an interactive Streamlit web application, seamlessly bridging data logic with accessible software to highlight communication and attention to detail

Automatic Crop Watering System

- Formulated an intelligent AI and IoT solution to optimize hydration schedules and minimize agricultural water waste, showcasing innovation and practical problem-solving.
- Programmed embedded hardware (C++/ Arduino) for sensor mapping and integrated a Python backend utilizing Blynk REST APIs and Gaussian Naive Bayes logic, demonstrating technical adaptability and cross-domain integration.

- Delivered a scalable dual-valve control system that operated physical relays and reduced test environment water usage by ~25%, highlighting analytical rigor and attention to sustainable impact

Solar Power Generation Forecasting

- Created a time-series forecasting model to predict solar power generation based on environmental data, optimizing energy resource allocation to showcase strategic planning and problem-solving.
- Implemented systematized data pipelines to process historical weather datasets, training machine learning regression algorithms in Python to demonstrate analytical thinking and technical adaptability.
- Validated system reliability by minimizing standard forecasting error metrics (RMSE/ MAE), delivering highly accurate temporal predictions that highlight analytical rigor and attention to detail.

Publications

AI and Sensor-Driven System for Irrigation and Water Waste Minimization,

Apr 2024

International Journal of Engineering Research & Technology

- Developed and published a research paper presenting an AI and IoT based irrigation management system that integrates soil moisture, temperature, and humidity sensors with machine learning models to optimize water usage.
- The system leverages Arduino and NodeMCU microcontrollers, cloud-based data processing, and mobile applications for real-time monitoring and computerized irrigation control.
- Demonstrated improvements in water conservation, crop yield optimization, and scalability for applications ranging from commercial farms to home gardens and greenhouses.

Education

MSc Mathematics, *Lovely Professional University*

Jul 2026 – Present | Punjab, India

Bachelor of Engineering in Computer Science and Engineering, *Nehru Institute of Engineering and Technology*
CGPA:8.01

Sep 2020 – Jul 2024 | Coimbatore, India

Certificates

Data Science, *ExcelR*

AWS Academy Machine Learning, *AWS Foundations*

AWS Academy Cloud, *AWS Foundations*

AI-ML Virtual Internship, *EduSkills*

CCNA: Introduction to Networks, *Cisco*